# Web Development and Web Design

Version Spring 2020

# Lab project

# Lab 2. Client-server interaction in scope of SPA

**Max 10 points.** *(Client-server interaction with pure JS, no jQuery. AJAX. Using Linter.)*
In scope of this lab students should practice fetching data from the back-end using AJAX.
Student shall create application which makes requests to API (back-end from the previous course can be reused) and displays parsed response.

Requirements:

- Pure JS, no jQuery or other libraries
- window.fetch or XMLHttpRequest can be used for AJAX calls
- Code should pass linter
- At least 3 API endpoints are used
- Application should be initialized as an npm project
- SPA shall communicate with back end via REST interface
- Data exchange shall be done in JSON format
- No errors nor logs in console
- Loading speed less than 4 seconds
- At least 2 browsers supported
- No commented code

**lint**, or a **linter**, is a tool that analyzes source code to flag programming errors, bugs, stylistic errors, and suspicious constructs.

Lint-like tools are especially useful for interpreted languages like **JavaScript** and Python. Because such languages lack a compiling phase that displays a list of errors prior to execution, the tools can also be used as simple debuggers for common errors (e.g. syntactic discrepancies) as well as hard-to-find errors such as heisenbugs (drawing attention to suspicious code as "possible errors"). Lint-like tools generally perform static analysis of source code.

Read more:

1. **Основи найпопулярнішого JavaScript лінтера для аналізу якості Вашого JS:** https://frontend-stuff.com/blog/eslint/
2. **Why you should always use a Linter** *by Alberto Gimeno*
medium.com/dailyjs/why-you-should-always-use-a-linter-and-or-pretty-formatter-bb5471115a76
3. **Thoughts about JavaScript linters and "lint driven development"** *by Daniel Sternlicht*
*https://medium.com/@danielsternlicht/thoughts-about-javascript-linters-and-lint-driven-development-7c8f17e7e1a0*

***Node.js & npm how-to:***

- Install latest node js and npm
  https://treehouse.github.io/installation-guides/windows/node-windows.html
  https://tecadmin.net/install-latest-nodejs-npm-on-ubuntu/
- Use npm to set up new npm package in application root folder
  https://docs.npmjs.com/cli/init

After previous steps student will have package.json configuration file and will be able to set up a linter.

***Linter configuration:***

- Add this packages to project devDependencies (package.json):
  - "eslint": "^6.8.0",
  - "eslint-config-airbnb-base": "^14.0.0",
  - "eslint-plugin-import": "^2.20.1"
- Add lint command to scripts section (package.json):
  - "lint": "eslint **/*.js --ignore-pattern node_modules/"
- Add .eslintrc.json file with such contents:

```
{
    "env": {
        "browser": true,
        "es6": true
    },
    "extends": [
        "airbnb-base"
    ],
    "globals": {
        "Atomics": "readonly",
        "SharedArrayBuffer": "readonly"
    },
    "parserOptions": {
        "ecmaVersion": 2018,
        "sourceType": "module"
    },
    "rules": {
       "indent": ["error", 4],
       "linebreak-style": "off"
    }
}
```

After completing the previous 3 steps students will be able to execute ‘npm run lint’ command to check codestyle etc.

**In the next parts:**
Lab 3. Basic single page application with user management. Framework. **Max 10 points**
Lab 4. Add project specific functionality to SPA. **Max 15 points**
Lab 5. Unit tests. **Max 15 points**
**Max 60 points for all 5 labs**